

Challenge #4

1. Access Dataset

I use Python, the programming language from the Google collaboratory platform, to manipulate data. Because the dataset is on kaggle.com, which is open source, we can directly access it from Google Colab with the following steps:

- Install 2 library :

```
!pip install opendatasets
!pip install pandas
```

- Import library

```
import opendatasets as od
import pandas as pd
```

- Read the dataset using the link source dataset from Kaggle

```
od.download("https://www.kaggle.com/datasets/datafiniti/womens-shoes-prices")
```

After the code is run, enter the Kaggle username and Kaggle key that can be obtained from the Kaggle account.

- Read datasets in the form of dataframes gar easier to manipulate

```
df = pd.read_csv("/content/womens-shoes-prices/Datafiniti_Womens_Shoes_Jun19.csv")
```

2. Data manipulation to get the answer

- What is the range of shoe prices?

Firstly, we filter our dataset only to retrieve information related to shoe products.

```
shoes = df[df["primaryCategories"]=="Shoes"]
```

Then, we can search for the price range of shoes by looking at the minimum and maximum price using min-max in Python.

```
print("minimum price : ", min(shoes["prices_amountMax"]))
print("maximum price : ", max(shoes["prices_amountMax"]))
```

The shoe price range was found to be 5.5 USD - 5000 USD.

- What are the median and average prices for women's shoes ?

```
median_price = shoes["prices_amountMax"].median()
average_price = shoes["prices_amountMax"].mean()
print("Median price:", median_price)
print("Average price:", average_price)
Obtained Median Price = 51.99 USD and Average Price = 72.45 USD
```

- How many unique shoe models does it contain?

```
category_counts = shoes['manufacturer'].value_counts()
print(category_counts)
category_counts = shoes['brand'].value_counts()
print(category_counts)
category_counts = shoes['colors'].value_counts()
print(category_counts)
```

In the shoe category in this dataset, if the uniqueness of shoes is seen from the manufacturer, there are 57 types; if seen from the brand, there are 505 different brands. Then, when viewed from the uniqueness of the colour, there are 1907 kinds of colours.

- How many different systems are shoe sizes encoded in this database?

```
category_counts = shoes['sizes'].value_counts()
print(category_counts)
491
```

- For how long has the database been maintained?
(as in, how many months are there between the first entry in the database and the latest entry?)
Because the date column dataset is still an object type, we need to change the dataset format to datetime format. After the data format is appropriate, we can find how long this database has been maintained by looking for the initial date the shoe data was entered and the last data entered in the database.

```
shoes['dateAdded'] = pd.to_datetime(shoes['dateAdded'])
shoes.sort_values('dateAdded')
min_date = shoes['dateAdded'].min()
max_date = shoes['dateAdded'].max()
duration_in_months = (max_date.year - min_date.year) * 12 + (max_date.month -
min_date.month)
print(duration_in_months)
```

The initial date of data entry was 2015-01-01, and the last date was 2019-01-01, so the database has been maintained for 48 months.

- How can this dataset be useful for the business problem at hand?
Of course, there are many things we can do with this dataset, including analysing what brands are selling the most, and then we can find what kind of shoe colour patterns are in demand the most. This insight can be an idea for shoe manufacturers with the most sales to produce shoes that suit market tastes and continue to dominate the market. This can also be an advantage for competitors to make shoes that can compete with shoe manufacturers who have been dominating the market.
- A unique insight or inspiration that you have taken from the dataset.
One insight I can gain from the data is that each shoe brand may have different market segments. This can be seen when I try to find the five brands with the most sales and the five brands with the highest prices.

```
category_counts = shoes['brand'].value_counts()
category_price = shoes[shoes['prices_amountMax']>4000]['brand'].value_counts()
top_5_brands_selling = category_counts.head(5)
print(top_5_brands_selling)
top_5_brands_expensive = category_price.head(5)
print(top_5_brands_expensive)
```

The five brands with the most sales are Brinley Co., SAS, Trotters, Propet, and Soda, while the five brands with the highest selling prices are Puma, Trotters, Softwalk, Stev-madden and Faux Suede. Four out of five have different segments. Some brands target high sales numbers and brands that target high selling prices. Usually, brands that target high sales use the concept of scarcity because the number of products produced is minimal, a 'limited edition', so the price is very high (information on the dataset is sold at 5000 USD).